Samarium

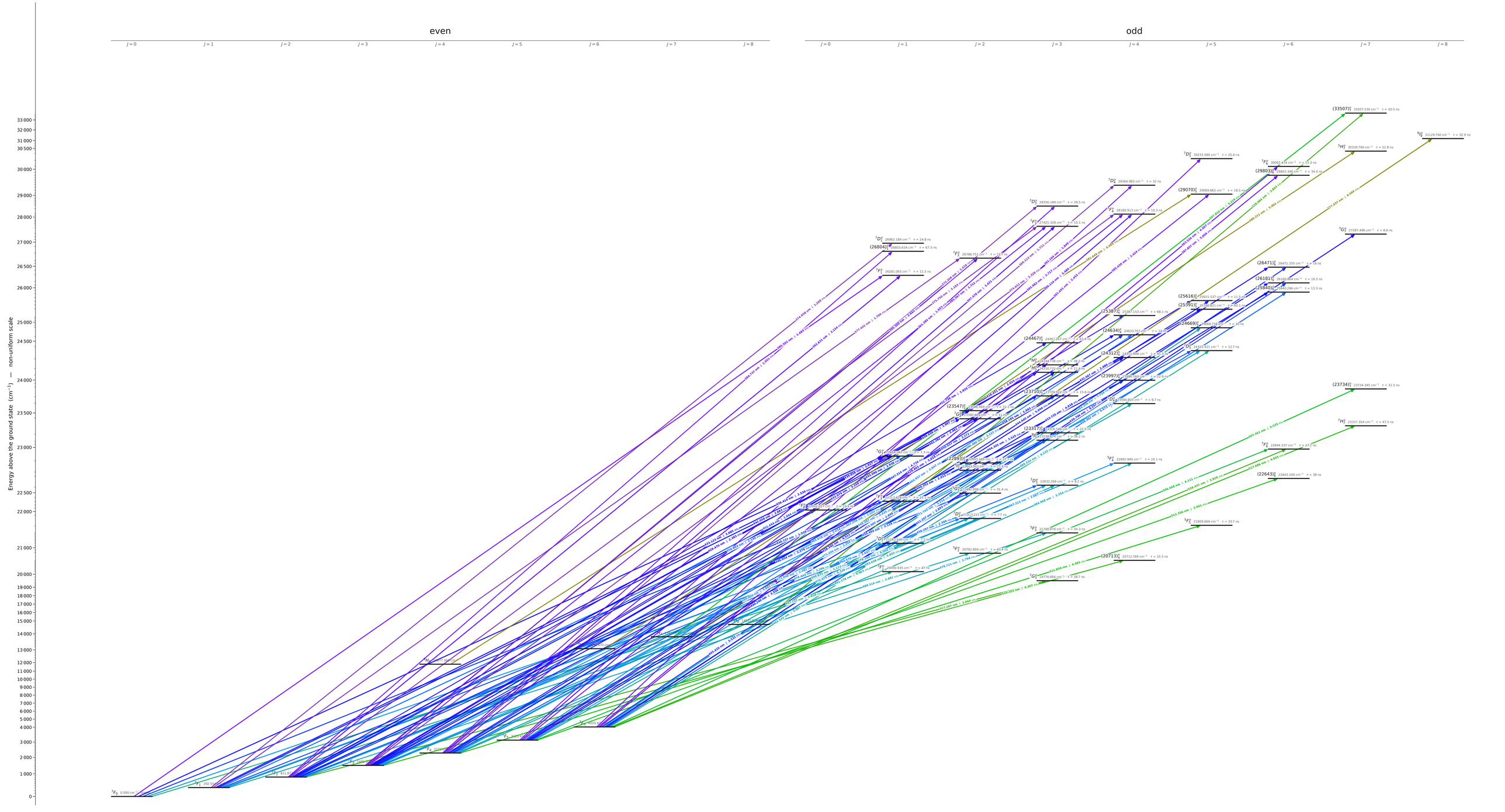
Sm (Sm I). Strongest classified transitions below 2 μm (NIST): 67 levels, 90 lines.

Sm⁺ ionisation limit 45519.69 cm^{-1} = 5.64372 eV (219.685 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000


even
odd
J = 0 J = 1 J = 2 J = 3 J = 4 J = 5 J = 6 J = 7 J = 8
Energy above the ground state (cm⁻¹) — non-uniform scale
0 1 000 2 000 3 000 4 000 5 000 6 000 7 000 8 000 9 000 10 000 11 000 12 000 13 000 14 000 15 000 16 000 17 000 18 000 19 000 20 000 21 000 22 000 22 500 23 000 23 500 24 000 24 500 25 000 25 500 26 000 26 500 27 000 28 000 29 000 30 000 30 500 31 000 32 000 33 000
$^7F_0$ 0.000 cm^{-1} τ = stable
$^7F_1$ 292.5 cm^{-1}
$^7F_2$ 611.8 cm^{-1}
$^7F_3$ 1060.7 cm^{-1}
$^7F_4$ 1654.8 cm^{-1}
$^7F_5$ 2415.8 cm^{-1}
$^7F_6$ 3369.5 cm^{-1}
$^7H_2$ 7600.9 cm^{-1}
$^7D_1$ 11961.4 cm^{-1}
$^7D_2$ 12705.0 cm^{-1}
$^7D_3$ 13598.0 cm^{-1}
$^7D_4$ 14620.7 cm^{-1}
$^7D_5$ 15748.3 cm^{-1}
$^7D_6$ 16960.0 cm^{-1}
$^7D_7$ 18230.0 cm^{-1}
$^7D_8$ 19530.0 cm^{-1}
$^7D_9$ 20830.0 cm^{-1}
$^7D_{10}$ 22100.0 cm^{-1}
$^7D_{11}$ 23300.0 cm^{-1}
$^7D_{12}$ 24400.0 cm^{-1}
$^7D_{13}$ 25300.0 cm^{-1}
$^7D_{14}$ 26000.0 cm^{-1}
$^7D_{15}$ 26500.0 cm^{-1}
$^7D_{16}$ 26800.0 cm^{-1}
$^7D_{17}$ 26900.0 cm^{-1}
$^7D_{18}$ 26900.0 cm^{-1}
$^7D_{19}$ 26900.0 cm^{-1}
$^7D_{20}$ 26900.0 cm^{-1}
$^7D_{21}$ 26900.0 cm^{-1}
$^7D_{22}$ 26900.0 cm^{-1}
$^7D_{23}$ 26900.0 cm^{-1}
$^7D_{24}$ 26900.0 cm^{-1}
$^7D_{25}$ 26900.0 cm^{-1}
$^7D_{26}$ 26900.0 cm^{-1}
$^7D_{27}$ 26900.0 cm^{-1}
$^7D_{28}$ 26900.0 cm^{-1}
$^7D_{29}$ 26900.0 cm^{-1}